THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS FULL SYLLABUS PAPER SET 03

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Which of the following points does not lie in Quadrant II? 1

(a) $(-7, 2)$ **(b)** $(4, -4)$ **(c)** $(-4, 4)$ **(d)** $(-1, 9)$

2. For the line $y = x + 3$, the slope and the y-intercept are respectively: 1

(a) 1 and 3 **(b)** 1 and -3
(c) 3 and 1 **(d)** -1 and 3

3. The number $0.333\ldots$ is: 1

(a) irrational **(b)** rational
(c) neither rational nor irrational **(d)** not a real number

4. Expand $(5x - 1)^2$: 1

(a) $25x^2 - 5x + 1$ **(b)** $25x^2 - 1$ **(c)** $25x^2 - 10x + 1$ **(d)** $25x^2 + 10x + 1$

5. The angle subtended by a diameter at any point on the circle is: 1

(a) 180° **(b)** 45° **(c)** 60° **(d)** 90°

6. ABCD is a cyclic quadrilateral with $\angle A = 65^\circ$. Then $\angle C$ is: 1

(a) 115° **(b)** 58° **(c)** 295° **(d)** 65°

7. Taking $\pi = \frac{22}{7}$, the area of a circle of radius 7 cm is: 1

(a) 154 cm^2 **(b)** 308 cm^2 **(c)** 44 cm^2 **(d)** 77 cm^2

8. An event occurred 6 times in 40 trials. Its experimental probability (relative frequency) is: 1

(a) 0.15 **(b)** 6.67 **(c)** 0.25 **(d)** 0.075

9. The probability of a certain event is: 1

(a) 0 **(b)** 100 **(c)** 0.5 **(d)** 1

10. The 10th triangular number is: 1

(a) 45 **(b)** 66 **(c)** 50 **(d)** 55

11. A triangle has sides 7 cm, 24 cm and 25 cm. By the converse of the Baudhāyana–Pythagoras theorem it is: 1

(a) right-angled, since all sides are different
(b) right-angled, since $a + b > c$
(c) not right-angled, since the sides are not equal
(d) right-angled, since $a^2 + b^2 = c^2$

12. The averages of 3 classes of 30 students each are 40, 52, 46. The overall average is: 1

(a) 47 **(b)** 46 **(c)** 45 **(d)** 48

13. Averaging the average heights of 8 seniors and 3 juniors directly does not give the group average because: 1

(a) the two groups have different sizes **(b)** the juniors are shorter

(c) the seniors are taller **(d)** averages cannot be combined

14. While adding two numbers digit by digit, the carry from any column can never exceed 1 because: 1

(a) the largest possible column total is 9 + 9 + 1 = 19

(b) the digits are always less than 5

(c) carry is always reset to 0

(d) the numbers are written in base 5

15. By Euclid's algorithm, gcd(1001, 143) is: 1

(a) 286 **(b)** 71 **(c)** 144 **(d)** 143

16. In $\triangle ABC$, P and Q are the midpoints of AB and AC and BC = 26 cm. The length of PQ is: 1

(a) 13 cm **(b)** 52 cm **(c)** 15 cm **(d)** 26 cm

17. The Varignon parallelogram of a quadrilateral whose diagonals are equal and perpendicular is: 1

(a) a square **(b)** a trapezium

(c) a rectangle that is not a square **(d)** a rhombus that is not a square

18. The graph of the equation x = 5 is a line that is: 1

(a) having slope 5 **(b)** parallel to the y-axis

(c) passing through the origin **(d)** parallel to the x-axis

19. The surface area of a sphere of radius r equals the curved surface area of a cylinder that tightly fits over it. The height of this cylinder is: 1

(a) πr **(b)** $4r$ **(c)** r **(d)** $2r$

20. The volume of a cuboid measuring 15 cm $\times$ 6 cm $\times$ 4 cm is: 1

(a) 25 cm^3 **(b)** 380 cm^3 **(c)** 360 cm^3 **(d)** 348 cm^3

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Write the coordinates of the point on the y-axis that is 6 units above the origin and of the point on the x-axis that is 4 units to the left of the origin. Find the distance between them. 2

22. Write the number of tiles in stage n if the pattern $1, 3, 5, 7, \ldots$ is followed, and find which stage has 47 tiles. 2

23. Show that $\frac{4}{6}$ and $\frac{10}{15}$ are equal rational numbers. 2

24. An isosceles triangle has equal sides 5 cm and base 6 cm. Find its area. 2

25. Explain with an example why finding one counterexample is enough to disprove a proposition. 2

26. Compute the combined percentage of profit when ₹2000 of sales give 20% profit and ₹3000 of sales give 30% profit. 2

27. A taxi charges a fixed amount ₹x plus ₹y per km. A 6 km ride costs ₹110 and a 10 km ride costs ₹170. Write a pair of linear equations for this. 2

28. How is a right circular cone obtained from a right-angled triangle? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each.

29. Find the perimeter of the triangle with vertices $(0,\,0)$, $(4,\,0)$ and $(4,\,3)$. 3

30. In $\triangle ABC$, M and N are the midpoints of AB and AC and D is any point on BC. Show that MN bisects AD. 3

31. Classify as rational or irrational: 3

(i) $\sqrt{81}$

(ii) $\sqrt{12}$

(iii) $0.\overline{3}$

(iv) $0.12345\,12345\ldots$ (repeating block). Give the fraction where rational.

32. Explain why three collinear points cannot lie on a circle. 3

33. Derive the formula for the sum of the first n natural numbers. 3

34. Line l_1 passes through $(0, 2)$ and $(3, 8)$; line l_2 passes through $(0, -4)$ and $(2, 0)$. Find the slope of each line and say whether the pair of equations has a solution. 3

35. State two situations from real life where the weighted mean is used and say what the weights are. 3

OR

Four hostel groups of 10 students each have average daily study times 2, 3, 4 and 3 hours. Find the overall average and state why simple averaging works here. 3

36. A wheel has diameter 56 cm. How many revolutions will it make to cover 1.76 km? ($\pi = \frac{22}{7}$) 3

OR

Find the perimeter of a sector of radius 21 cm and angle 60° ($\pi = \frac{22}{7}$). 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. **Case study:** A shop sells two combo packs. Pack 1 has 2 pens and 3 notebooks for ₹80; Pack 2 has 3 pens and 2 notebooks for ₹70. 5

(i) Form the pair of equations in the pen price x and notebook price y.

(ii) Solve it.

(iii) What is the cost of 1 pen and 1 notebook together?

(iv) What does a unique solution mean here?

OR

Prove that the slope of $y = mx + d$ is m using any two points on the line, and use it to find the slope and y-intercept of $6x - 3y + 9 = 0$. 5

38. A trader says: “If I sell more articles, my profit is more.” 5

(i) Write the converse.

(ii) A shop sells 10 pens at ₹4 profit each and then 20 pens at ₹1 profit each; show that selling more gave less profit.

(iii) Give a situation to show that the converse can be false.

(iv) What are such examples called?

OR

(i) Show that “If n is a product of two unequal primes, then it has exactly 4 divisors” is true using 15 and 33. 5

(ii) Is the converse true? Find a counterexample if any.

39. If $x + \frac{1}{x} = 5$, find $x^2 + \frac{1}{x^2}$ and $x^3 + \frac{1}{x^3}$. 5

OR

Factorise 5

(i) $x^2 - 5x + 6$,

(ii) $r^2 - r - 42$,

(iii) $6x^2 + 7x + 2$,

(iv) $10x^2 - 11x - 6$.

40. **Case study:** A class of 50 students gives its mode of travel to school: bus 18, bicycle 10, walking 12, car 10. 5

(i) Find the probability that a randomly chosen student cycles.

(ii) Find the probability for each mode and check that they add to 1.

(iii) Among 1200 students in the school, estimate the number who walk.

(iv) Which mode is most likely and by how much more than the least likely?

OR

Case study: In a factory, 8 of a sample of 200 bulbs are defective. 5

(i) Find the probability that a bulb is defective.

(ii) Estimate the number of defective bulbs in a batch of 5000.

(iii) Find the probability that a bulb is good.

(iv) Why would a bigger sample give a more reliable estimate?

END OF QUESTION PAPER **Total: 80 Marks**

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